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Location of Hamstring Injuries Based on Magnetic Resonance Imaging: A Systematic Review

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Context: Hamstring muscle injury location using magnetic resonance imaging (MRI) is not so well described in the literature.

Objective: To describe the location of hamstring injuries using MRI.

Data Sources: PubMed, Web of Science, Scopus, SPORTDiscus, Cochrane Library.

Study Selection: The full text of studies, in English, had to be available. Case reports and reviews were excluded. Included studies must report the location of hamstring injuries using MRI within 8 days of the acute injury.

Study Design: Systematic review.

Level of Evidence: Level 4.

Data Extraction: A first screening was conducted based on title and abstract of the articles. In the second screening, the full text of the remaining articles was evaluated for the fulfillment of the inclusion criteria.

Results: From the 2788 references initially found in 5 databases, we included 34 studies, reporting a total of 2761 acute hamstring injuries. The most frequent muscle head involved was the long head of the biceps femoris (BFLH) (70%), followed by the semitendinosus (ST) (15%), generally associated with BFLH. The most frequent tissue affected was the myotendinous junction (MTJ) accounting for half the cases (52%). Among all lesions, the distribution between proximal, central, and distal lesions looked homogenous, with 34.0%, 33.4% and 32.6%, respectively. The stretching mechanism, while only reported in 2 articles, represented 3% of all reported mechanisms, appears responsible for a specific lesion involving the proximal tendon of the semimembranosus (SM), and leading to a longer time out from sport.

Conclusion: BFLH was the most often affected hamstring injuries and MTJ was the most affected tissue. In addition, the distal, central, and proximal locations were homogeneously distributed. We also noted that MRI descriptions of hamstring injuries are often poor and did not take full advantage of the MRI strengths.

Systematic Review Registration: Before study initiation, the study was registered in the PROSPERO International prospective register of systematic reviews (registration number CRD42018107580).

Keywords: hamstring; muscle; strain; tear; location; MRI; sports injury

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In high-speed running sports, hamstring muscles are the most frequently injured, leading to serious consequences for athletes (eg, time loss, performance decrease, recurrence, financial loss).[15-17,20,56] Despite efforts to prevent and manage hamstring injuries,[4,5,30,35,44] their incidence has not decreased.[19,20] One explanation could be that these measures have not sufficiently focused on the particularities of the hamstring muscle group, which consists in 4 muscles of the posterior thigh: long head of the biceps femoris muscle (BFLH), short head of the biceps femoris muscle (BFSH), semitendinosus muscle (ST), and semimembranosus muscle (SM).[49] These 4 muscles differ in their anatomy, structure, and function.[6,29,49] BFLH and SM are hemipennate muscles, ST has a fusiform shape and BFSH has a slanted trapezoid shape.[29] The fiber length per total muscle length of BFLH and SM are smaller than those of ST and BFSH.[29] They are biarticular: hip extensor and knee flexor muscles (except uniarticular BFSH).[6] Their involvement measured via electromyography or functional magnetic resonance imaging (MRI) differed according to different exercises and activities.[7,34] The injury locations also varied according to sports, with a reported predominance of BFLH injuries for sprinting sports (eg, football)[11] and of SM for stretching sports (eg, dance),[1,2] supporting the different biomechanical role of each hamstring muscle. The hamstring muscles are composed of different tissues (ie, tendon, myotendinous junction (MTJ), muscle, aponeurosis, myoaponeurosis junction) with varying degrees of resistance to breakage.[6] The involved tissue could play a role in the healing process of the injury, and thus in the duration of time loss in sport or injury recurrence.[41] The different size of the injury (eg, area, volume, height) could also be a relevant characteristic of the injury, given its influence on the healing process and time-loss duration.[2]

Consequently, "hamstring injury" does not appear as a unique injury, but more as a generic term including several injuries resulting from the combination of a specific muscle and tissue location, which has been reported to have specific biomechanical and healing properties.[6,29,49] Each specific hamstring injury may require its own optimal management in prevention or rehabilitation, suggesting the need for a clear and detailed hamstring injury diagnosis including at least location and tissue involved.[30,41] It thus seems essential to explore further the specific hamstring injuries. MRI can be used to provide an accurate and reliable description of hamstring injury location and tissue involved,[1,9,48,50,54] since MRI presents a very high sensitivity and seems not to be operator dependent.[54] A clear understanding on the different hamstring injuries would be of help to improve hamstring injury prevention and rehabilitation programs by targeting the most frequent injury locations and also the variety of hamstring injuries. Through this systematic review, we aimed to specify anatomic and tissue locations of acute hamstring injuries using early MRI to provide an in-depth analysis of hamstring injuries.

METHODS

Our systematic review was registered in the PROSPERO database (registration number: CRD42018107580).

The literature search was performed on March 1, 2020, in the following 5 databases: PubMed/MEDLINE, SPORTDiscus, Scopus, Web of Science, and the Cochrane Library, to identify studies dealing with hamstring injuries using a combination of key words: injuries or its synonyms (*lesion*, *wounds*, *pain*, *tear*, *damage*, *strain*, *laceration*, *rip*, *rent*, *split*, *rupture*, *pred*, *caus*, *risk factors*) AND imaging or synonyms (*MRI*, *magnetic resonance imaging*) AND hamstring or its synonyms (*hamstring muscles*, *posterior thigh*, *biceps femoris*, *semitendinosus*, *semimembranosus*) (keywords equations in Appendix 1, available in the online version of this article).

Studies were included if (1) MRIs were performed within 8 days of an acute hamstring injury: we chose this delay since no changes in MRI signal have been reported within this time frame, especially in edema and fiber disruption extent,[52] and less is known about the MRI signal after 8 days (there is a risk not to detect the lesion or underestimate the lesion size given the edema resorption); (2) MRI findings were used as a descriptive and/or diagnostic tool for hamstring injury; (3) studies reported injury location as seen on MRI; (4) injury mechanism was indirect; (5) the article was written in English; and (6) the full text of the article was available. Interventional studies were included for (1) those testing a prevention program only if the outcomes regarding the hamstring injury location seen on MRI could be extracted and for the intervention group if the MRI was performed before the intervention and (2) those with an injection we only considered the initial MRI performed after injury but before any injection. All narrative, systematic, or meta-analysis reviews and case reports were excluded.

Two researchers independently screened studies based on title and abstract according to the inclusion/exclusion criteria, using Covidence (https://www.covidence.org). Then, the same 2 researchers reviewed the selected full texts. Discrepancies were resolved by consensus. If no consensus could be reached, a third reviewer was available for a final decision, but this was not required. Studies referenced in the selected articles were also screened for relevant studies not already highlighted by our search.

One researcher assessed the potential risk of bias of the studies included, using the criteria of the consensus statement of Hayden et al[25] including 14 items. A score ≥75% was required to consider a low risk of bias of the study.

One researcher extracted the following data: author and year of publication; study design; patients' follow-up; studied population (number, sex, age, sport and level, country, history of hamstring injury); MRI properties (magnetic field used, MRI sequences); number of hamstring injuries; number of injuries by location: muscle head (long or short head of the biceps femoris, semitendinosus, semimembranosus), part of the muscular

system affected (tendon/aponeurosis, muscle belly, MTJ), proximal, central, distal, or extensive (combining the 3) location, as well as the classification used to describe injury location (eg, British Athletics Muscle Injury classification[39]); number of injuries according to the grade of severity, as well as the classification used to describe injury severity (eg, classification of Peetrons[40]); size of lesions: height, length, width, median area, average volume, distance from ischial tuberosity, average length of the tendinous lesion, percentage of tendon area affected, measure of tendinous retraction; and mechanisms of injuries. In general, the volume of the lesions was estimated by multiplying the height of the lesion by its anteroposterior diameter and by the width, dividing by 2 (the lesion being considered as an ellipse).[48]

The "primary lesion" was defined as the muscle lesion with the largest volume of injury. The "secondary lesion" was defined as the muscle lesion with the second largest volume of injury. The area involved was defined as the largest cross-sectional area of the muscle head seen on axial view. Professional athletes or those competing at national or international level were considered as "high level," whereas amateur, recreational athletes, or those competing at regional or lower levels of competition, were considered as "low level."

To provide a synthesis of the results from included studies, a descriptive analysis was performed.

RESULTS

The flowchart of the selection of articles is presented in Figure 1. We included 34 articles. Their description is presented in Table 1, and a summary of their characteristics is presented in Table 2. Regarding the risk of bias assessment, all the studies met the quality criteria (score ≥75%): one study had a score of 79%,[14] and the other 33 had scores of 93% or 100% (Appendix 2 available online). The duration of the participants' follow-up in studies varied from 19 days to 10 years (Appendix 3 available online).

Regarding MRI methodology, most MRIs were performed at a magnetic field strength of 1.5 T units (n = 28, 82.4%) or 3.0 T (n = 4, 11.8%), and 2 used a 1.0 T (n = 2, 5.9%) magnetic field. Protocols included T2-weighted sequences with fat signal suppression obtained with spectral suppression (fat saturation pulse [n = 13, 43.3%] inversion recovery [short tau IR] [n = 9, 30.0%]), or without fat signal suppression (n = 7, 23.3%), proton density (n = 11, 36.6%), and T1 (n = 18, 60.0%) sequences. The sequences contained at least 2 planes per inclusion criteria in 31 articles, with a maximum of 3 for 6 articles. All the MRIs were performed within 8 days of the hamstring injury occurring, with an average of 4.3 ± 1.6 days. Of the 34 included studies, only 23 used a classification for injury location (Appendix 4 available online). Six different classifications were described, such as BAMI, Chan, Peetrons, classic (rupture with more than or less than 50%), tear/edema/avulsion, or a new classification. Among the 23 studies (corresponding to 1822 hamstring injuries[8,10,14,17,18,21-23,32,33,37,42-46,50-55]) using a classification for injury severity, the Peetrons's being the most frequent (65% of studies) (Table 2).

Combining all the 34 included studies, a total of 2761 hamstring injuries were reported. An overall view of the characteristics of these hamstring injuries is presented in Figure 2 and Table 2. Moreover, 32 (94.1%) studies mentioned the muscle head involved, and 18 (52.9%) described the tissue location. A description of tissue location within each muscle head was given in only 2 studies (5.9% of included studies). The frequency of hamstring injuries features, ranged from 15.6% to 82.8% of the number injuries, is described in Appendix 4 (available online).

Combining the included studies, more than 1 muscle head was affected in 25.4% of the hamstring injuries. In 0.9% of the injuries, 3 muscle heads were affected. All the injuries (including primary and secondary lesions) were described in 20 studies, for a total of 1068 hamstring injuries:[1,2,4,8,9,13,21,28,32,37,38,42,44-46,48,50,52,53,55] BFLH represented 69.8% of total injuries, ST 15.4%, SM 13.1%, and BFSH 1.7%. Primary lesions were described in 30 studies, for a total of 2438 hamstring injuries:[1,2,4,5,10-14,17,18,21-23,28,32,33,37,38,42-46,48,50-53,55] with lesions of BFLH largely predominant (76.9%) (Figure 2 and Table 2). When 2 muscles were affected, BFLH (primary lesion) and ST (secondary lesion) was the most frequent combination (82.1%), others were less common (BFLH and BFSH, BFLH and SM, ST and SM, with 7.7%, 3.8%, and 6.4%, respectively).

Regarding the type of tissue affected (described in 18 studies, for 1286 hamstring injuries[1,1,2,4,5,9,11-13,18,28,38,42,43,48,52-55]), the MTJ was involved in more than half the cases (n = 668, 51.9%) (Figure 2 and Table 2).

The proximal-central-distal location was described in only 40.6% of the studies. When considering distal and proximal locations (described in 7 studies, for 713 hamstring injuries[1,3,11,13,32,37,55]), the muscular lesions were mostly proximal (62.8%) (Figure 2 and Table 2). When considering central and extensive lesions (described in 8 studies, for 344 hamstring injuries[8,23,38,42,43,50,52,54]), the distribution looked homogenous, with 34.0%, 33.4%, and 32.6% for proximal, central, and distal locations, respectively.

The description combining both muscle head and locations within the muscle was made in only 2 studies[11,13] dealing with 416 lesions[11] and 15 lesions,[13] respectively. Detailed results of hamstring injury distribution are presented in Appendix 5 (available online).

Sizing presented by dimension and area was provided in 20 and 16 studies (58.8 and 47.1%), respectively, and the volume in 10 studies (29.4%), and summary of mean sizes is presented in Figure 2 and Table 2.

Regarding the severity, grade 1 (which differed slightly between classification systems) was the most frequently reported (n = 923, 50.7%), followed by grade 2 (n = 713, 39.1%), grade 0 (n = 147, 8.1%), and grade 3 (n = 39, 2.1%).

About a quarter of patients presented a history of hamstring injury (n = 279 of 1086 lesions, 25.7%) but only 2 studies[28,55] provided data regarding the location of previous injuries and

Table 1. Description of the 34 articles included in the present systematic review[a]

Study	Study Design	Country	No. of Participants (Total [Men/Women]) Age, y: Mean (SD-Min-Max)	Sport	Level[b] (No. of Patients)	Classification Used	No. of Lesions	Delay Injury/MRI, days: Mean (SD-Min-Max)	Muscle Concerned (BFLH/BFSH/ST/SM), %	Primary Localization Inside Muscle (MTJ/MAJ/Aponeurosis/Muscle/Tendon), %
Askling et al[2]	Prospective	Sweden	15 (1/14) NR (NR-16-24)	Dance	High (15)	NR	13	4	0/0/0/100	0/0/0/0/100
Askling et al[1]	Prospective	Sweden	18 (10/8) NR (NR-15-28)	Athletics (Sprint)	High (18)	NR	26	4	100/0/0/0	89/0/0/11/0
Askling et al[5]	Prospective	Sweden	75 (69/6) 25 (5-NR-NR)	American football	High (67) and low (8)	NR	75	5	76/NR/0/24	71/NR/NR/NR/NR
Askling et al[4]	Prospective	Sweden	56 (36/20) 20 (NR-NR-NR)	Athletics (sprint and jump)	High (56)	NR	56	5	79/0/14/7	43/NR/NR/NR/NR
Cohen et al[8]	Retrospective	USA	38 (38/0) 27 (NR-22-35)	American football	High (38)	1 = nothing; 2 = ≤50% rupture; 3 = 50% rupture; new classification proposed	55	3	NR/NR/NR/NR/NR	NR/NR/NR/NR/NR
Connell et al[9]	Prospective	Australia	60 (60/0) 24 (4-NR-NR)	American football	High (60)	NR	42	3	NR/NR/NR/NR/NR	55/37/0/0/8
Crema et al[11]	Retrospective	Qatar	275 (275/0) 25 (5-18-39)	American football	High (275)	Edema vs tear vs avulsion	393	5	56/6/24/14	52/16/NR/32/NR
Crema et al[10]	Retrospective	Brazil	22 (22/0) 26 (5-19-24)	Soccer	High (22)	Modified Peetrons	22	3	66/0/28/6	63/37/0/0/0
Crema et al[12]	Retrospective	USA	11274 (NR) NR (NR-NR-NR)	Several	High (38)	Modified Peetrons/BAMI	29	7[c]	100/0/0/0	NR/NR/NR/NR/NR
Desmet et al[13]	Retrospective	USA	15 (14/1) NR (NR-18-22)	Several	Low (15)	NR	15	5	73/0/20/2	47/NR/NR/53/NR

(continued)

Table 1. (continued)

Study	Study Design	Country	No. of Participants (Total [Men/Women]) Age, y: Mean (SD-Min-Max)	Sport	Level[b] (No. of Patients)	Classification Used	No. of Lesions	Delay Injury/MRI, days: Mean (SD-Min-Max)	Muscle Concerned (BFLH/BFSH/ST/SM), %	Primary Localization Inside Muscle (MTJ/MAJ/Aponeurosis/Muscle/Tendon), %
Devos et al[14]	Prospective	New Zealand	161 (61/3) 28 (NR-23-33)	Several	High (49) and low (15)	Edema vs tear vs avulsion	64	5	88/NR/NR/NR	NR/NR/NR/NR/NR
Ekstrand et al[17]	Prospective	Sweden	816 (207/0) NR (NR-NR-NR)	Soccer	High (207)	Modified Peetrons	207	2	NR/NR/NR/NR	NR/NR/NR/NR/NR
Ekstrand et al[18]	Prospective	Sweden	255 (255/0) NR (NR-NR-NR)	Soccer	High (255)	Modified Peetrons	255	2	84/NR/5/11	61/25/4/10/NR/
Gibbs et al[21]	Prospective	Australia	13 (13/0) NR (NR-18-33)	Australian football	High (13)	Edema vs tear vs avulsion	17	3	58/NR/21/21	NR/NR/NR/NR/NR
Hallen et al[22]	Prospective	Sweden	249 (249/0) NR (NR-NR-NR)	Soccer	High (249)	Modified Peetrons	249	2	83/NR/5/12	NR/NR/NR/NR/NR
Hamilton et al[24]	Prospective	Qatar	25 (NR) NR (NR-NR-NR)	Several	High (24) and low (1)	Modified Peetrons	25	5	NR/NR/NR/NR	NR/NR/NR/NR/NR
Hamilton et al[23]	Prospective	Qatar	139 (139/0) 26 (NR-18-39)	Soccer	High (139)	New classification proposed	110	5	81/1/3/15	NR/NR/NR/NR/NR
Koulouris et al[28]	Prospective	Australia	41 (41/0) NR (NR-NR-NR)	Australian football	High (41)	NR	51	3	88/NR/4/8	53/41/NR/NR/6
Nercolarde et al[37]	Prospective	Spain	7 (7/0) 19 (NR-NR-NR)	Soccer	High (7)	Edema vs tear vs avulsion	7	1	71/NR/0/29	NR/NR/NR/NR/NR
Opar et al[38]	Prospective	Australia	99 (99/0) NR (NR-NR-NR)	Australian football	High (99)	NR	17	3	76/0/12/12	70/0/NR/24/6
Pollock et al[42]	Retrospective	England	44 (28/16) 24 (4-18-39)	Athletics	High (44)	BAMI	65	7	70/3/10/17	50/16/NR/NR/

(continued)

Table 1. (continued)

Study	Study Design	Country	No. of Participants (Total [Men/Women]) Age, y: Mean (SD-Min-Max)	Sport	Level[b] (No. of Patients)	Classification Used	No. of Lesions	Delay Injury/MRI, days: Mean (SD-Min-Max)	Muscle Concerned (BFLH/BFSH/ST/SM), %	Primary Localization Inside Muscle (MTJ/MAJ/Aponeurosis/Muscle/Tendon), %
Pomeranz et al[43]	Retrospective	USA	12 (12/0) NR(NR-20-29)	Athletics	High (12)	1 = nothing; 2 = ≤50% rupture; 3 = 50% rupture	12	8	50/NR/8/42	44/NR/NR/44/12
Rettig et al[44]	Retrospective	USA	(53/0) 25 (NR-22-28)	American football	High (10)	1 = edema = 8 cm or inferior; 2 = >8 cm; rupture	10	2	80/NR/0/20	NR/NR/NR/NR/NR
Reunik et al[18]	Prospective	Netherlands	53 27 (NR-18-46)	Several	High (24) and low (29)	Modified Peetrons	53	5	80/0/4/16	NR/NR/NR/NR/NR
Reunik et al[45]	Prospective	Netherlands	108 (108/3) 28 (7-NR-NR)	Several	High (44) and low (64)	Modified Peetrons	108	5	81/NR/4/15	NR/NR/NR/NR/NR
Slavotinek et al[48]	Prospective	Australia	37 (37/0) 24 (NR-17-32)	Australian football	High (37)	New classification proposed	30	6	70/0/30/0	93/7/NR/NR/NR
Vandermade et al[32]	Prospective	Netherlands	165 (NR) 26 (NR-NR-NR)	Several	High (87) and low (68)	Modified Peetrons	165	5	82/NR/4/14	0/0/0/0/100
Vandermade et al[33]	Prospective	Netherlands	70 (70/0) 24 (NR-21-30)	Several	High (69) and low (1)	Modified Peetrons/BAMI	70	5	80/NR/3/17	0/0/0/0/100
Verral et al[50]	Prospective	Australia	83 (83/0) NR (NR-NR-NR)	Australian football	High (83)	New classification proposed	80	6	72/NR/21/7	NR/NR/NR/NR/NR

(continued)

Table 1. (continued)

Study	Study Design	Country	No. of Participants (Total [Men/Women]) Age, y: Mean (SD-Min-Max)	Sport	Level[b] (No. of Patients)	Classification Used	No. of Lesions	Delay Injury/MRI, days: Mean (SD-Min-Max)	Muscle Concerned (BFLH/BFSH/ST/SM), %	Primary Localization Inside Muscle (MTJ/MAJ/Aponeurosis/Muscle/Tendon), %
Wangensteen et al[51]	Prospective	Qatar	180 (180/0) 26 (5-NR-NR)	Several	High (177) and low (3)	Modified Peetrons	141	5	79/1/3/17	NR/NR/NR/NR/NR
Wangensteen et al[55]	Prospective	Qatar	180 (180/0) 26 (5-NR-NR)	Soccer	NR	Modified Peetrons	19	5	87/0/3/10	79/NR/NR/8/13
Wangensteen et al[52]	Prospective	Qatar	132 (132/0) 31 (NR-20-49)	Several	High (7) and low (5)	Modified Peetrons	12	1	92/0/0/8	55/NR/NR/9/36
Wangensteen et al[54]	Prospective	Qatar	40 (40/0) 26 (NR-19-46)	Several	High (39) and low (1)	Modified Peetrons/BAMI/Chan	56	5	NR/NR/NR/NR	86/NR/NR/9/5
Wangensteen et al[53]	Prospective	Qatar	176 (176/0) 26 (5-NR-NR)	Several	High (173) and low (3)	Modified Peetrons/BAMI/Chan	212	5	80/1/4/15	23/26/NR/NR51

BAMI, British Athletics Muscle Injury classification; BFLH, long head of the biceps femoris; BFSH, short head of the biceps femoris; MAJ, myoaponeurotic junction; MRI, magnetic resonance imaging; MTJ, myotendinous junction; NR, not reported; SM, semimembranosus; ST, semitendinosus muscle.
[a]Some studies may have involved identical or partially identical populations, but there was no way to differentiate them, so this remains a hypothesis, which is why they were all included separately.
[b]Professional athletes or those competing at national or international level were considered as "high level" while amateur, recreational athletes, or those competing at regional or lower levels of competition, were considered as "low level."
[c]Not strictly precise but considered as 1 week maximum (Olympic Games).

Table 2. Summary of the characteristics of the 34 studies included and of the 2761 hamstring injuries included

Study design, n (%)	
Retrospective	8 (23.5)
Prospective	26 (76.5)
Injured population	
Age, y, mean ± SD (min-max)	25.6 ± 5.2 (18.8-36.7)
Sex, n (%)	
Total	2773 (100)
Men	2702 (97.4)
Women	71 (2.6)
Prior HMI, n (%)	
Yes	279 (25.7)
No	807 (74.3)
Sports, n (%)	
Several sports	12 (35.3)
Soccer	9 (26.5)
Australian football	5 (14.7)
Track and field	4 (11.8)
American football	3 (8.8)
Dancers	1 (2.9)
Country, n (%)	
Qatar	8 (23.5)
Sweden	7 (20.6)
Australia	6 (17.6)
USA	5 (14.7)
Netherlands	4 (11.8)
Brazil	1 (2.9)
New Zealand	1 (2.9)
Spain	1 (2.9)
England	1 (2.9)
Level, n (%)	
Low	426 (16.7)
High	2124 (83.3)

(continued)

Table 2. (continued)

Hamstring injuries	
No. of injuries	2761
Injury size, mean ± SD	
Height, cm	14.4 ± 2.6
Length, cm	2.4 ± 0.7
Width, cm	2.4 ± 0.8
Area, cm^2	26.8 ± 10.6
Volume, cm^3	89.8 ± 27.0
Distance to ischial tuberosity	10.7 ± 5.2
Injury localization, %	
Musculotendinosus junction	51.9
Musculoaponeurotic junction	17.9
Tendon	16.0
Muscle	13.4
Fascia	0.8
Injury position, %	
Proximal	34.0
Central	33.4
Distal	32.6
Mechanism, n (%)	
Sprinting	436 (55.3)
Stretching	24 (3.0)
Nonsprinting	196 (24.9)
Sprinting + stretching	131 (16.6)
Undefined	1 (0.1)
Primary lesion, n (%)	
Semimembranosus	13.4
Semitendinosus	8.6
Long head of the biceps femoris	79.6
Short head of the biceps femoris	1.1

BAMI, British Athletics Muscle Injury classification; HMI, hamstring muscle injuries.

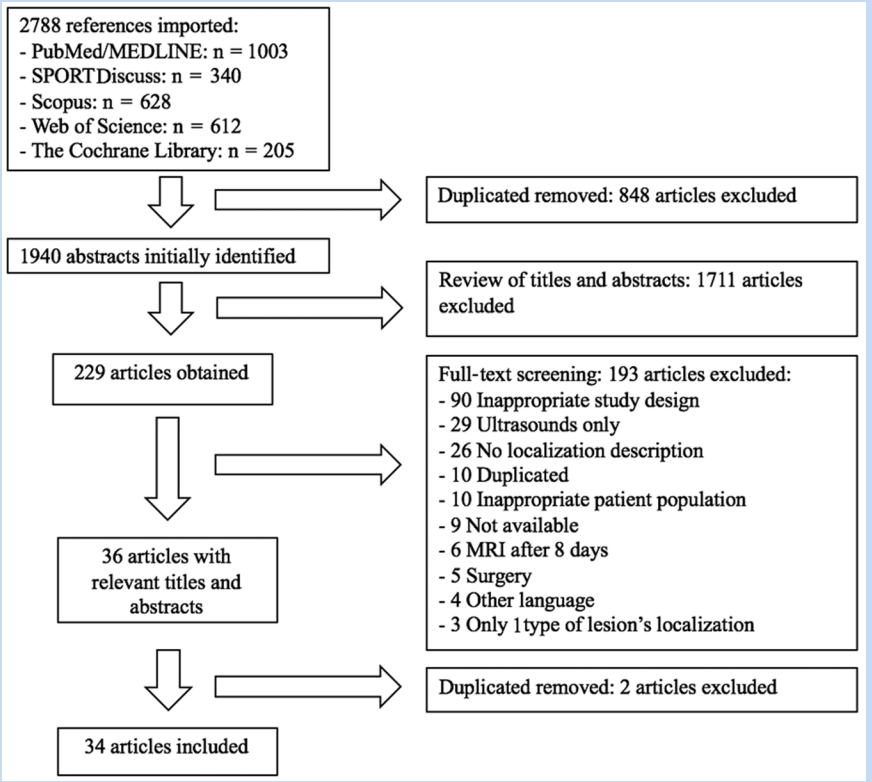
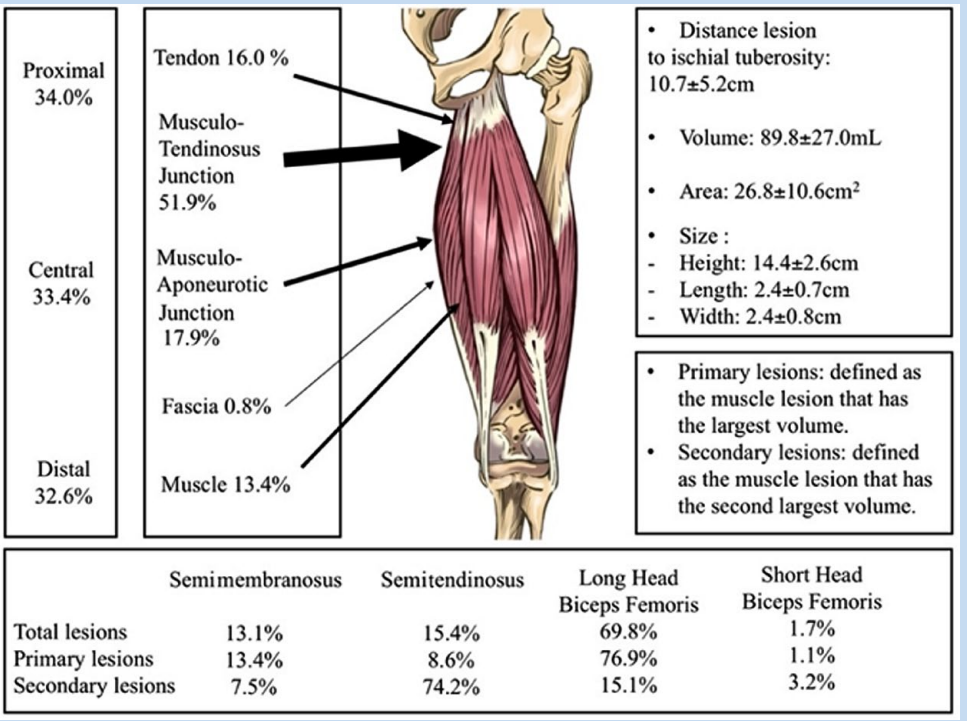


Figure 1. Flowchart illustrating the different stages of the study inclusion and exclusion process in the present systematic review.



	Semimembranosus	Semitendinosus	Long Head Biceps Femoris	Short Head Biceps Femoris
Total lesions	13.1%	15.4%	69.8%	1.7%
Primary lesions	13.4%	8.6%	76.9%	1.1%
Secondary lesions	7.5%	74.2%	15.1%	3.2%

Figure 2. Location of hamstring injuries based on 2761 injuries detailed in the 34 included studies.

reinjury risk. Of the 29 hamstring reinjuries in these 2 studies, 24 (82.8%) concerned BFLH, 1 (3.4%) ST, and 4 (13.8%) SM. The affected tissues were for 18 (62.1%) MTJ, 5 (17.2%) MAJ (myoaponeurotic junction), 2 the muscle belly (6.9%), and 4 (13.8%) the tendon. In only 1 study[55] was the grade of reinjury specified, and of these 19 reinjuries, 10 (52.6%) were grade 1, 7 (36.8%) were grade 2, and 2 (10.5%) were grade 3.

Regarding the clinical mechanism (reported in 9 studies and 788 hamstring injuries[1,2,4,5,14,17,38,51,53]), the majority (n = 436, 55.3%) occurred during sprinting activities. Of the 24 hamstring injuries caused by maximal stretching, 15 were reported in a study[2] with a patient population of dancers, and 14 concerned the SM, and 13 the proximal part of the tendon. An undefined mechanism (n = 1, 0.1%) was rarely described.

DISCUSSION

The main findings of the present study were that hamstring injuries most frequently involved BFLH (69.8% of all lesions and 76.9% of primary lesions) and its MTJ and are equally likely to be seen in its proximal, central, or distal areas. In addition, our results support the fact that there is currently no agreement/consensus about classification to be used.

Our findings confirm earlier studies reporting that BFLH was the most frequently injured hamstring muscle,[27,39] with a more accurate view of the problem. Although some hypotheses have been put forward to explain the higher susceptibility of hamstring muscles to injury (eg, biarticular,[29] composed mainly of fast fibers,[49] subject to significant eccentric forces during explosive sprint and jump performances,[26] and/or peak myotendinous strain and negative work during the late swing phase or heel strike[26]), the fact that BFLH is most often involved is not well explained. In a quarter of the cases, more than 1 muscle was injured. This condition may have been underestimated because some articles never mention a second muscle affected and it is not possible to know if this is because there was no secondary injury or if they only reported the main injury. BF and ST was the most common combination probably because of their anatomic proximity.

Although the size of the lesions was described in less than 50% of the lesions, it can be seen that the shape was very ovoid, with a height on average more than 5 times greater than the anteroposterior diameter and the transverse diameter. The lesion thus seems to extend in general in the direction of the shape of the muscle.

The MTJ was involved in more than half of cases (51.9%), far ahead of the other locations. The MTJ of all the hamstring musculature is very extensive: It runs the entire length of the muscle from the proximal free tendon to the distal free tendon except for the BFLH muscle, which has 2 intramuscular tendons that traverse most but not all of the muscle and slightly overlap (1 medial and 1 lateral).[49] The reason for the predisposition of the MTJ continues to be sought.[31] Nevertheless, the area of attachment between the muscle and the tendon could be an area of weakness, and probably explains this high frequency of injury compared with other structures. The other explanation would simply be to the fact that the 2 MTJs (proximal and distal) are globally quite extensive and poorly defined spatially (often difficult to differentiate them from aponeurotic junctions), and therefore these lesions would often be so classified by the radiologist.

An important finding of our systematic review was the scarcity and heterogeneity of the description of hamstring injuries reported in the selected literature. Some articles only described 1 or 2 parameters, ignoring parameters such as muscle heads, grade, tissue, clinical mechanisms, precise dimensions, sex, sport, history of injuries (eg, anterior cruciate ligament rupture), prior hamstring injuries, or chronic phenomena. Only 2 articles described the muscle heads and tissues involved in the hamstring injuries, but even these have their limitations. The first study[11] did not consider the lesions of tendon or fascia separately and some data seem to be different from that generally reported in the literature, whereas the second study[13] dealt with only 15 lesions. There is a clear need for detailed descriptive studies of hamstring muscle injuries.

The strengths of this systematic review are the rigorous search (2788 articles analyzed from 5 databases) with a low risk of methodological bias of this systematic review. Moreover, the study of the 34 articles retained for analysis provided us a consequent number of hamstring injuries (n = 2761), which allows us to give some weight to our conclusions of the distribution of hamstring muscle injuries according to muscle head, tissue, and location. In addition, all MRIs were performed within 8 days of the injury, which gives additional strength to the study because the MRI signal did not change in this interval. Beyond that, there would be a risk of not detecting the lesion or underestimating the size or grade of the lesion by resorption of edema.[52] Given the results of the risk of bias assessment, we consider that all the studies included demonstrate low risk of bias, which enhances the quality of the results obtained from this systematic review.

We have to acknowledge some limitations of our systematic review. Despite a rigorous systematic search, we may have missed articles. The restriction on English language was based on findings from systematic reviews suggesting no evidence of bias for conventional medicine if studies written in languages other than English were excluded.[36] The risk of methodological bias for each study seems to be very low, even if potential confounders are possible. Since we are interested in the location and not in the prognosis, we have freed ourselves from the biases previously described on the articles of the hamstrings (especially the fact that clinicians are not blinded to MRI data).[47] In addition, even if systematic reviews are exposed to bias given by MRI interpretation from different radiologists, we know that there is a very good intra- and interobserver reliability, including MRI data analysis of muscle injuries.[54] Nevertheless, and as mentioned before, the heterogeneity of the available data is an intrinsically limiting factor. Even if some data are detailed in each article, there is no complete description of these lesions.

As perspectives we can suggest that the use of MRI data needs to be further developed, to be more homogenous, and

consensus must be reached so that we all use the same terms to describe these injuries. New MR sequences, including quantitative ones (eg, fat, edema, and spectroscopy) may help us better understand the differences between the various muscle heads. Moreover, injuries in women are very insufficiently represented in the literature.

CONCLUSION

This study provides a clear view of the distribution of hamstring injuries analyzed by MRI based on a very high number of injuries. The distribution of the hamstring injury location is inhomogeneous, with a high predisposition of BFLH and MTJ. Some injuries seem more associated with a particular injury mechanism, such as SM proximal involvement with or without tendon avulsions. Thus, hamstring injury should be considered as specific lesions, and on well-detailed MRI diagnosis can help to design specific injury-based rehabilitation and prevention programs.

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